biotechnology, nanotechnology, military, automobile... the list goes on. These hybrid metals are stronger and more durable than the base metals that go into producing them, not to mention way more cost effective.

Without the invention of alloys we would be left deficient in many areas. The invention of the Airplane which also features in this list would have been impossible as there are no naturally occurring metals that can withstand the immense pressures and stresses of high altitudes. With no corrosion resistant stainless steel electronic, agricultural, road and rail industries would have to continually replace resources. We'd find ourselves with large holes in our pockets from having to buy gold in its pure 24-karat form if we didn't have cheaper options courtesy the invention of the alloy.

Amplifiers

Not the amplifiers that give you a headache and make you call the police on your neighbours. These amplifier circuits are quite clever techniques of increasing the signal strength that allow for communication between a tiny microprocessor and the world. They're used at the beginning and end points of all electronic systems to ensure reliable communication. Amplifiers strengthen the power of a signal that needs to be processed by other elements down the electronic circuit. The tiny command from the microprocessor that has a current value of microamperes goes through an amplifier and then to the rest of the circuit. Nothing too complicated,